QISKIT FALL FEST 2026 / TRACK 2

Portfolio selection with QAOA

Constraint-preserving search, warm starts, and the cost of hardware realism



TEAM AXIOM

Abhiram Arun N S V · Sree Sruthi Alur

Main result: classical baselines reach the exact optimum.

SIMULATOR STUDY • NO QPU JOBS

A portfolio is a binary decision under a hard budget

Choose exactly K assets from n candidates

$$C(x) = q\, x^T \Sigma x - \mu^T x + \lambda(\sum_i x_i - K)^2$$

$x_i \in \{0,1\}$; $x_i = 1$ means asset i is selected.

Σ captures pairwise risk; μ represents expected return.

INSTANCE

n = 12 assets • K = 6

q = 0.5 • data seed = 2026

SEARCH SPACE

C(12,6) = 924 feasible portfolios

Exact enumeration is the reference.

EXACT INTERMEDIATE OPTIMUM

C* = −1.066392

Assets [1, 5, 6, 8, 10, 11]

Encoding: $x_i = (I - Z_i)/2$. Penalties discourage violations; a constraint-preserving mixer avoids them by design.

03

Three QAOA designs test different ways to handle feasibility

A / XY-QAOA

INITIAL STATE

Uniform superposition over weight-6 bitstrings

MIXER

Ring XY interactions preserve Hamming weight

TRADE-OFF

Feasibility = 1 by construction; costly circuit.

B / Warm start

INITIAL STATE

Product state from continuous relaxation

MIXER

Standard X mixer + penalty $\lambda = 1$

TRADE-OFF

Best observed optimal-solution probability.

C / Penalized QAOA

INITIAL STATE

$|+\rangle^n$, no problem-specific bias

MIXER

Standard X mixer + penalty $\lambda = 1$

TRADE-OFF

Simple baseline; feasibility not guaranteed.

COMMON SETUP

Depth p = 1–3 • COBYLA multistart • 4,096 exact-statevector samples • simulator-only

Compared with exhaustive enumeration and classical greedy / simulated annealing.

Penalty strength matters: too little penalty rewards invalid portfolios

Beginner instance: n = 6, K = 3

Only 20 portfolios satisfy the constraint.

All 64 bitstrings can be enumerated exactly.

EXACT PENALTY THRESHOLD

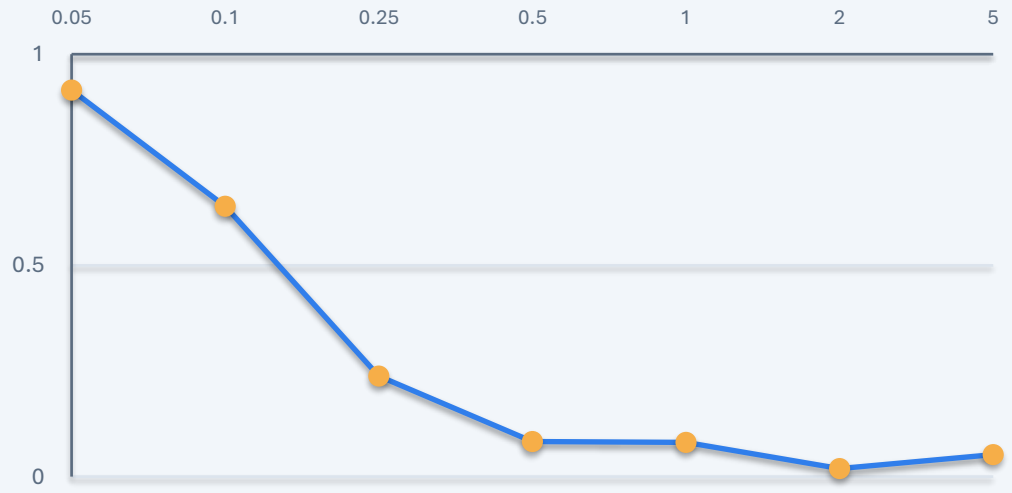
λ* = 0.109661226

Below λ*, an infeasible state can win globally.

REFERENCE OPTIMUM

C* = −0.503014690 • assets [1, 2, 5]

Observed feasible-sample fraction vs λ



Best sampled feasible cost matched the exact optimum at every tested depth.

05

Warm start improves optimal-hit probability, not best achievable cost

Method	Feasibility (p=3)	P(opt) (p=3)	Best feasible cost	Interpretation
A · XY mixer	100%	0.0066	−1.066392	**Feasibility**
B · Warm start	69.1%	0.1609	−1.066392	**Best P(opt)**
C · Penalty	60.6%	0.0007	−1.066392	**Baseline**
Greedy / SA	100%	Exact optimum	−1.066392	**Classical wins**

Greedy and feasible-swap simulated annealing found the exact optimum in under 2 seconds.

Relaxation + rounding also recovered the optimum; continuous relaxation value: −1.120144.

06

Routing and state preparation dominate the circuit budget

TARGET

FakeBrisbane model
127 qubits · heavy-hex

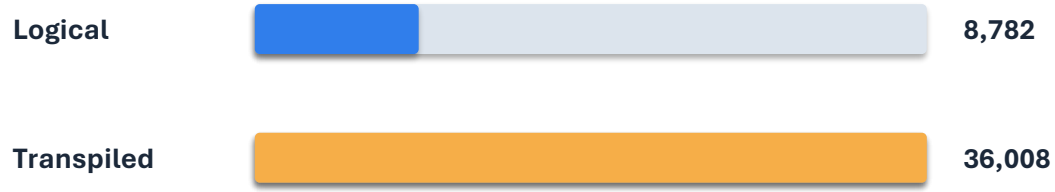
LOGICAL CIRCUIT

Depth 8,782
4,623 two-qubit gates

TRANSPILED SUBGRAPH

Depth 36,008
10,554 ECR gates

Circuit depth



WHERE THE COST COMES FROM

Custom weight-K state preparation accounts for 92.7% of logical depth and 8

51.4% of logical two-qubit pairs are non-adjacent, increasing routing cost.

Overhead: 4.10× depth and 2.28× two-qubit count.
Compiled estimates only; no real QPU run.

The gate-based circuit differs from exact XY evolution before noise is added

1 / IMPLEMENTATION GAP

Exact weighted-subspace evolution vs noiseless Trotterized circuit

TV = 0.5827
P(opt): 0.00664 exact → 0.00211 circuit

2 / NOISE DEGRADATION

FakeBrisbane-derived noisy simulation

Feasibility: 20.7% (1,024 shots)
No optimal bitstring observed in that sample.

Distribution distance (total variation)



INTERPRETATION

Trotterization first; backend-derived noise on top.

FakeBrisbane is a backend-derived noise model, not a physical IBM device.

What the experiments actually establish

01	**No advantage demonstrated**	Classical greedy, SA, and relaxation + rounding reach the exact n=12 optimum.
02	**Constraint preservation has a price**	XY-QAOA guarantees exactly six selected assets, but the compiled circuit is extremely deep.
03	**Warm start is the strongest sampler here**	Method B reaches P(opt) ≈ 0.161 at p=3, yet does not beat the classical optimum.
04	**Separate the error sources**	The noiseless Trotter circuit has an implementation gap; backend-derived noise degrades it further.

One synthetic instance • exact classical reference • simulator-only • no quantum-advantage claim

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